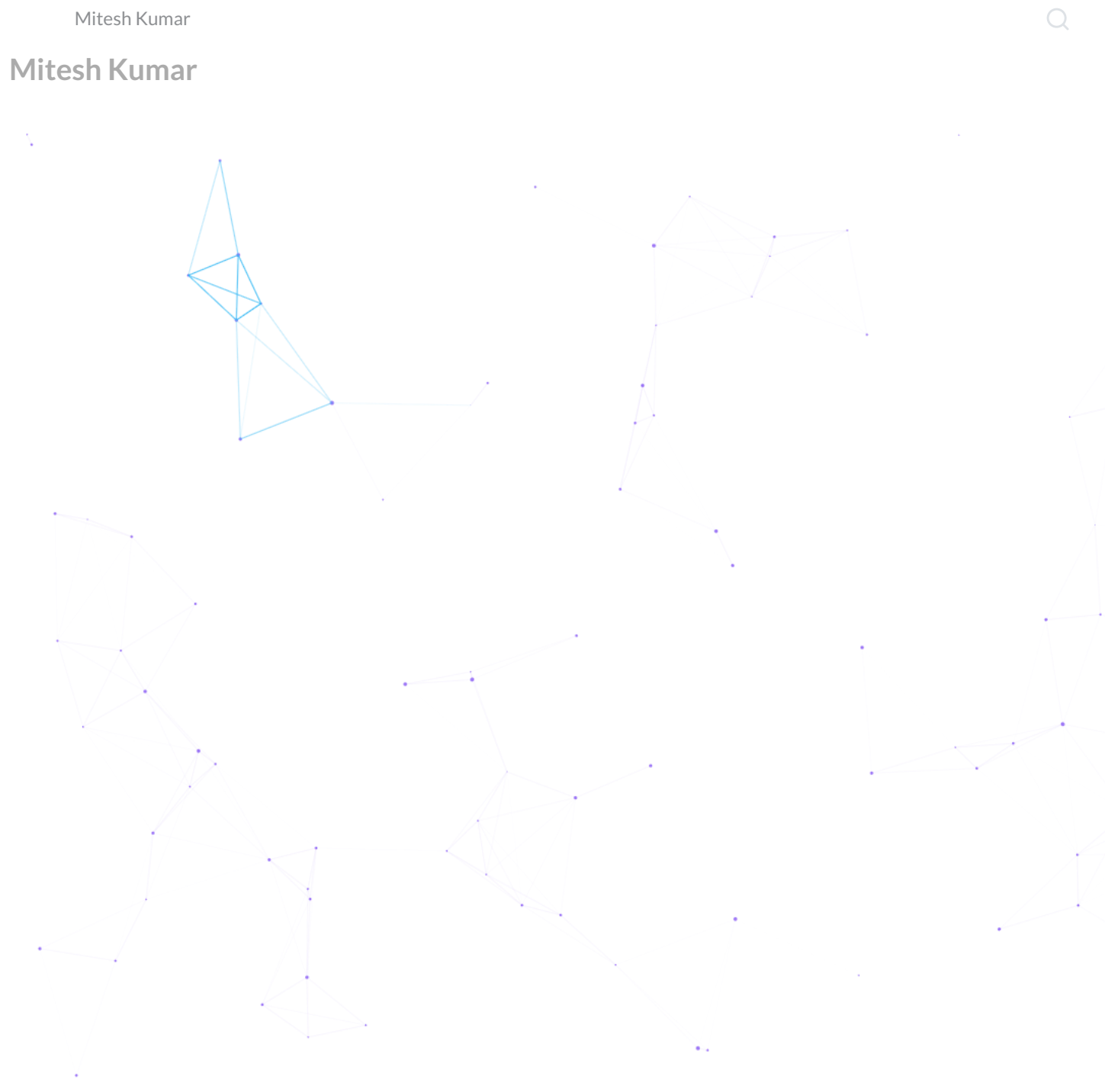




PORTFOLIO // 2026

MITESH KUMAR

DATA SCIENTIST & ML ENGINEER

Specializing in **Mathematical Optimization**, **Computer Vision**, and **Deep Learning**. I architect robust AI pipelines and solve complex network systems at scale.

[Explore Work](#) [View Resume](#)

Experience

Dec 2025 – Present

Student Research Assistant
University of Regensburg

Machine Learning for Side-Channel Analysis. Reviewing deep learning-based side-channel attacks and investigating robustness of neural-network-based attacks under realistic measurement conditions.

Deep Learning Security
May 2025 – Present

Student Research Assistant

FAU Erlangen-Nürnberg

Transregio 154: Engineered a bicriteria optimization model in Pyomo for gas networks. Scaled numerical experiments across HPC clusters using Slurm job arrays.

Pyomo Gurobi HPC

Featured Work

Computer Vision

Open-Set Face Recognition & Robust Tracking

Built a robust face recognition pipeline achieving 90% accuracy in open-set identification. Optimized efficiency with dynamic K-Means clustering and dual-stream FaceNet (ResNet50) for 128-dimensional embeddings.

Integrated MTCNN for detection with Template Matching fallback tracker for identity continuity during occlusions.

FaceNet MTCNN ResNet50
Object Detection

Hierarchical Object Detection

Engineered an object detection pipeline integrating multi-scale Selective Search, ResNet18 CNN, and Linear SVM. Achieved MABO of 0.74 via NMS.

SVM CNN

Computational Imaging

Synthesized HDR images by reverse-engineering camera response functions and Bayer demosaicing.

3D Scene Perception

Extracted 3D dimensions from noisy ToF sensor data using MLESAC plane-fitting.

Technical Arsenal

Languages

Python SQL C++

ML Frameworks

PyTorch TensorFlow Scikit-learn

Optimization

Pyomo Gurobi Pandas